

STATUS: 100%
OPERATIONAL

ALL 13 TESTS
PASSED

LIVE MVP
COMPLETE

Report Generated: September
2026

StagePilot (StageFlow) – Project Progress Report

Real-Time Event Orchestration, Cascading Delay Engine & AI Anchor Teleprompter

Project Implementation Completion

100% (Sprint Objectives Met)

1. COMPLETED FEATURE MILESTONES

✓ Cascading Schedule Delay Engine

Allows organizers to add +5m, +10m, or +15m. Automatically cascades and shifts all downstream agenda start times without page refreshes.

✓ Full Stage Teleprompter

High-contrast teleprompter with auto-scroll speed controls and 1-click horizontal mirror reflection mode for physical beamsplitter glass.

✓ Phonetic Pronunciation Guide

Prevents speaker mispronunciations by highlighting phonetic guides prominently on cue cards (e.g., [eh-LEH-nah ross-TOH-vah]).

✓ Emergency Flash Alerts & Chime

Instant backstage broadcast popups for stage anchors with pure synthesized 800Hz → 600Hz audio chime via Web Audio API.

✓ Offline-Resilient AI Co-Pilot

Dynamic context-assembling prompt generator for speaker intros and delay banter, with seamless Smart Fallback when offline.

✓ Zero-Config Embedded Database

Integrated MongoMemoryServer fallback that boots and auto-seeds the tech summit conference if no local MongoDB is installed.

2. AUTOMATED VERIFICATION & TEST RESULTS

Verification Area	Target Component / Endpoint	Result	Notes
Service Health	GET /api/health	PASSED	Service reported online with zero downtime
Authentication	POST /api/auth/login (Organizer)	PASSED	JWT token issued; role verified as ORGANIZER
Authentication	POST /api/auth/login (Anchor)	PASSED	JWT token issued; role verified as ANCHOR
Agenda Ingestion	GET /api/events/:id	PASSED	Loaded 4 sessions with populated speaker records
Pronunciation Data	Speaker.pronunciationGuide	PASSED	Phonetic spellings verified on speakers
Real-Time Sync	Socket.IO (join_event)	PASSED	Bi-directional room connection established

Verification Area	Target Component / Endpoint	Result	Notes
Delay Engine	POST /api/events/:id/sessions/:id/delay	PASSED	+10m delay calculated; event health updated
Flash Broadcast	POST /api/events/:id/broadcast	PASSED	Broadcast received live across WebSocket clients
AI Script Engine	POST /api/ai/generate-script	PASSED	Contextual delay script generated dynamically
AI Stage Copilot	POST /api/ai/copilot-query	PASSED	Instant stage icebreaker returned in under 1s

3. LIVE SERVICE ENDPOINTS & QUICK ACCESS

Service	URL	Credentials / Role
Frontend Web App	http://localhost:5173	Active (Vite Dev Server)
Backend API & WebSockets	http://localhost:5000	Active (Express + Socket.IO)
Organizer War-Room	http://localhost:5173/organizer	organizer@stagepilot.io / password123
Stage Anchor Teleprompter	http://localhost:5173/anchor	anchor@stagepilot.io / password123

4. NEXT PHASE ROADMAP (FUTURE SCOPE)

**Speech-to-Text Teleprompter Follower**

Highlight words in real time as the anchor speaks and synchronize auto-scroll speed directly to vocal cadence.

**Multi-Track / Multi-Stage Support**

Simultaneous management of multiple parallel stages (Auditorium A, Workshop Hall B, Demo Stage C).

**Audience Interactive Q&A**

Live attendee QR-code question submissions and upvoting pushed directly to the anchor's cue card.

**Run-of-Show PDF & Calendar Sync**

1-click export of the live adjusted schedule into printable cue sheets and CalDAV / Google Calendar feeds.